

Performance Evaluation of Machine Learning Algorithms

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Outline

- ▶ Introduction
- ▶ Predictive performance Measures
 - ▶ Classification
 - ▶ Regression
- ▶ Performance Evaluation Methods
 - ▶ Holdout
 - ▶ Repeated holdout
 - ▶ Cross-validation
 - ▶ Bootstrap
- ▶ Methods for model comparisons
 - ▶ Significance test

Performance Evaluation

- ▶ Performance of a machine learning algorithm can be evaluated in the following aspects
 - ▶ Predictive performance
 - ▶ How accurate is the learned model in prediction?
 - ▶ Interpretability
 - ▶ Complexity of the learned model
 - ▶ Efficiency
 - ▶ Time complexity (speed)
 - ▶ Time to build the model
 - ▶ Time to classify examples using the model
 - ▶ Scalability
 - ▶ How run time changes with the increase of size of data.
 - ▶ Memory complexity
- ▶ Here we focus on the **predictive performance**

Predictive Performance Evaluation

- ▶ Objective:
 - ▶ Find out how good is a learned *model* (i.e., classifier or regressor) in making predictions on *a test set of examples*?
 - ▶ The test set should be different from the training data from which the model is built from
- ▶ Performance measures depend on the task
 - ▶ Classification
 - ▶ Regression

Performance Measures for Classification

- ▶ Performance measures
 - ▶ Classification accuracy
 - ▶ Classification error rate
 - ▶ Classification cost
 - ▶ Precision
 - ▶ Recall
 - ▶ F-measure
 - ▶ Area Under ROC Curve (AUC)
 - ▶ etc.

Accuracy and Error Rate

- ▶ **Accuracy** of a classifier on a data set:

$$\frac{\text{number of examples classified correctly}}{\text{total number of examples in the data set}}$$

- ▶ **Error rate** of a classifier (= 1 – accuracy)

$$\frac{\text{number of examples classified incorrectly}}{\text{total number of examples in the data set}}$$

- ▶ Accuracy or error on the training data is not a good indicator of the classifier's performance on future data. Should use accuracy or error on a test data set.

Confusion Matrix

- ▶ Confusion matrix
 - ▶ Records the numbers of correct and incorrect classifications of a classifier on a test data set.
 - ▶ Is often used to calculate all the metrics
- ▶ For problems with 2 classes:

		Predicted class		
		Class = yes	Class = no	Total
Actual class	Class = yes	# of yes examples classified into yes	# of yes examples classified into no	# of true yes examples
	Class = no	# of no examples classified into yes	# of no examples classified into no	# of true no examples
	Total	# of predicted yes examples	# of predicted no examples	Total # of examples

Confusion Matrix

- Expressed using TP, FP, FN and TN:

- ▶ TP: true positive
- ▶ FP: false positive
- ▶ FN: false negative
- ▶ TN: true negative

Predicted class

Actual class		Class = yes	Class = no	Total
	Class = yes	TP	FN	TP+FN
	Class = no	FP	TN	FP+TN
	Total	TP+FP	FN+TN	TP+TN+FP+FN

- ▶
$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

- ▶
$$\text{Error rate} = \frac{FP + FN}{TP + TN + FP + FN}$$

Example

- ▶ Consider the following **confusion matrix**:

		Predicted class		
Actual class		cancer = yes	cancer = no	Total
	cancer = yes	90	210	300
	cancer = no	140	9560	9700
	Total	230	9770	10000

- ▶ What is the accuracy?
 - ▶ $\text{Accuracy} = (90+9560)/10000 = 96.50\%$
- ▶ What is the error rate?
 - ▶ $\text{Error rate} = (140+210)/10000 = 3.5\%$

Limitation of Accuracy or Error Rate

- ▶ Consider a 2-class data set:
 - ▶ Number of Class 0 examples = 9,990
 - ▶ Number of Class 1 examples = 10 (e.g., fraud cases in fraud detection)
- ▶ If a model predicts everything to be class 0, what is the accuracy?
 - ▶ $9990/10000 = 99.9 \%$
- ▶ Accuracy is misleading because model does not detect any class 1 example (which is often important)
 - ▶ Do not consider the cost of different types of misclassification

Misclassification Cost

- ▶ Misclassification cost

- ▶ In practice, different types of misclassifications often incur different costs.

- ▶ E.g., in making loan decisions, the cost of lending to a defaulter is far greater than the lost-business cost of refusing a loan to a non-defaulter.

- ▶ Cost matrix:

		Predicted class	
		Class 1	Class 2
Actual class	Class 1	0	Cost of misclassifying a Class 1 example to Class 2
	Class 2	Cost of misclassifying a Class 2 example to Class 1	0

Misclassification Cost (*Cont'd*)

- ▶ Calculating misclassification cost of a model on a test set

- ▶ Confusion matrix: **Predicted class**

Actual class		Class 1	Class 2
	Class 1	# of Class 1 examples classified into Class 1	# of Class 1 examples classified into Class 2
	Class 2	# of Class 2 examples classified into Class 1	# of Class 2 examples classified into Class 2

- ▶ Misclassification cost: $\sum_i \text{cost}_i \times \text{num}_i$

where cost_i is the cost in the i th cell of the cost matrix and num_i is the value in the i th cell of the confusion matrix.

Classification Cost (*Cont'd*)

► Example

► Cost matrix:

		Predicted class	
		Class 1	Class 2
Actual class	Class 1	0	6
	Class 2	1	0

► Confusion matrix of models on a test data set:

from model 1:

Actual class	Predicted class	
	Class 1	Class 2
Class 1	65	10
Class 2	20	40

from model 2:

Actual class	Predicted class	
	Class 1	Class 2
Class 1	70	5
Class 2	30	30

► Misclassification cost:

80

What about accuracy?

60

Performance Measures for Classification

- ▶ Performance measures
 - ▶ Classification accuracy
 - ▶ Classification error rate
 - ▶ Classification cost
 - ▶ Precision
 - ▶ Recall
 - ▶ F-measures
 - ▶ Area Under ROC Curve (AUC)
 - ▶ etc.

Precision, Recall and F-measure

- ▶ Usually for measuring the performance on predicting examples of one class (the interesting, usually small class)
- ▶ **Precision:** (Consider that examples of the class in question are positive)
 - ▶ Exactness: what % of examples that the classifier labeled as positive are actually positive

$$precision = \frac{TP}{TP + FP}$$

- ▶ **Recall**

- ▶ Completeness - what % of actual positive examples are classified as positive

$$recall = \frac{TP}{TP + FN}$$

		Predicted class		
Actual class		Class = yes	Class = no	Total
	Class = yes	TP	FN	TP+FN
	Class = no	FP	TN	FP+TN
	Total	TP+FP	FN+TN	TP+TN+FP+FN

Precision, Recall and F-measure

- ▶ F measure (F1 or F-score): harmonic mean of precision and recall

$$F_score = \frac{2 \times precision \times recall}{precision + recall}$$

Precision, Recall and F-measure

- ▶ For measuring performance on all the classes in the data set, take an average among all classes
 - ▶ Macro-average
 - ▶ Take the average of precisions or recalls on all classes
 - ▶ Treat each class equally
 - ▶ Used when classes are imbalanced
 - ▶ Micro-average
 - ▶ Sum up the individual true positives, false positives, and false negatives of the system for different sets and
 - ▶ Use them to compute overall precision and recall.
 - ▶ Cares the model's overall performance instead of per-class performance
 - ▶ Used when classes are balanced
- ▶ F-score is computed the same way

Performance Measures for Classification

- ▶ Performance measures
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ROC Curve and AUC

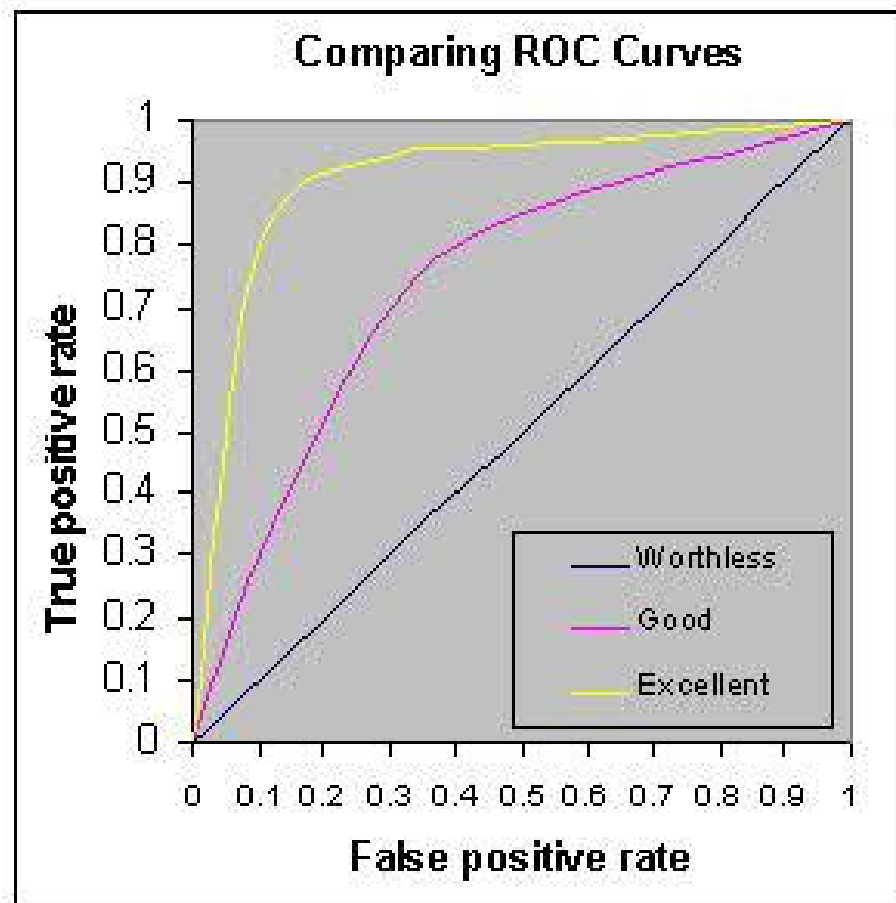
- ▶ ROC = Receiver Operating Characteristic
 - ▶ Started in electronic signal detection theory (1940s - 1950s)
 - ▶ Has become very popular in machine learning applications to assess classifiers
- ▶ ROC curve: a plot of true positive rate (TPR) against false positive rate (FPR)

		Predicted class		
		Class = yes	Class = no	Total
Actual class	Class = yes	TP	FN	TP+FN
	Class = no	FP	TN	FP+TN
	Total	TP+FP	FN+TN	TP+TN+FP+FN

$$TPR = \frac{TP}{TP+FN} = Recall \qquad FPR = \frac{FP}{FP + TN}$$

ROC Curve

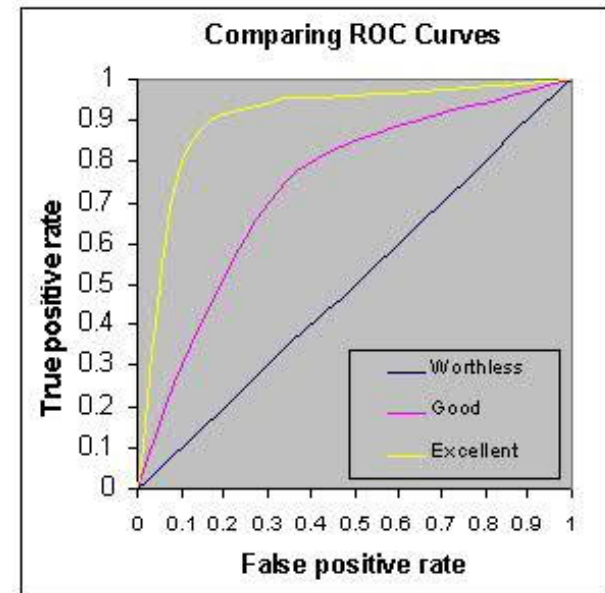
- Plot of TPR against FPR



How to Obtain ROC Curve for a Classifier

- ▶ Given a test data set, there is usually one point (FPR, TPR) in the ROC space - How to create a curve?
- ▶ ROC is often used for a binary classifier that outputs the probability that an example belongs to the positive class.

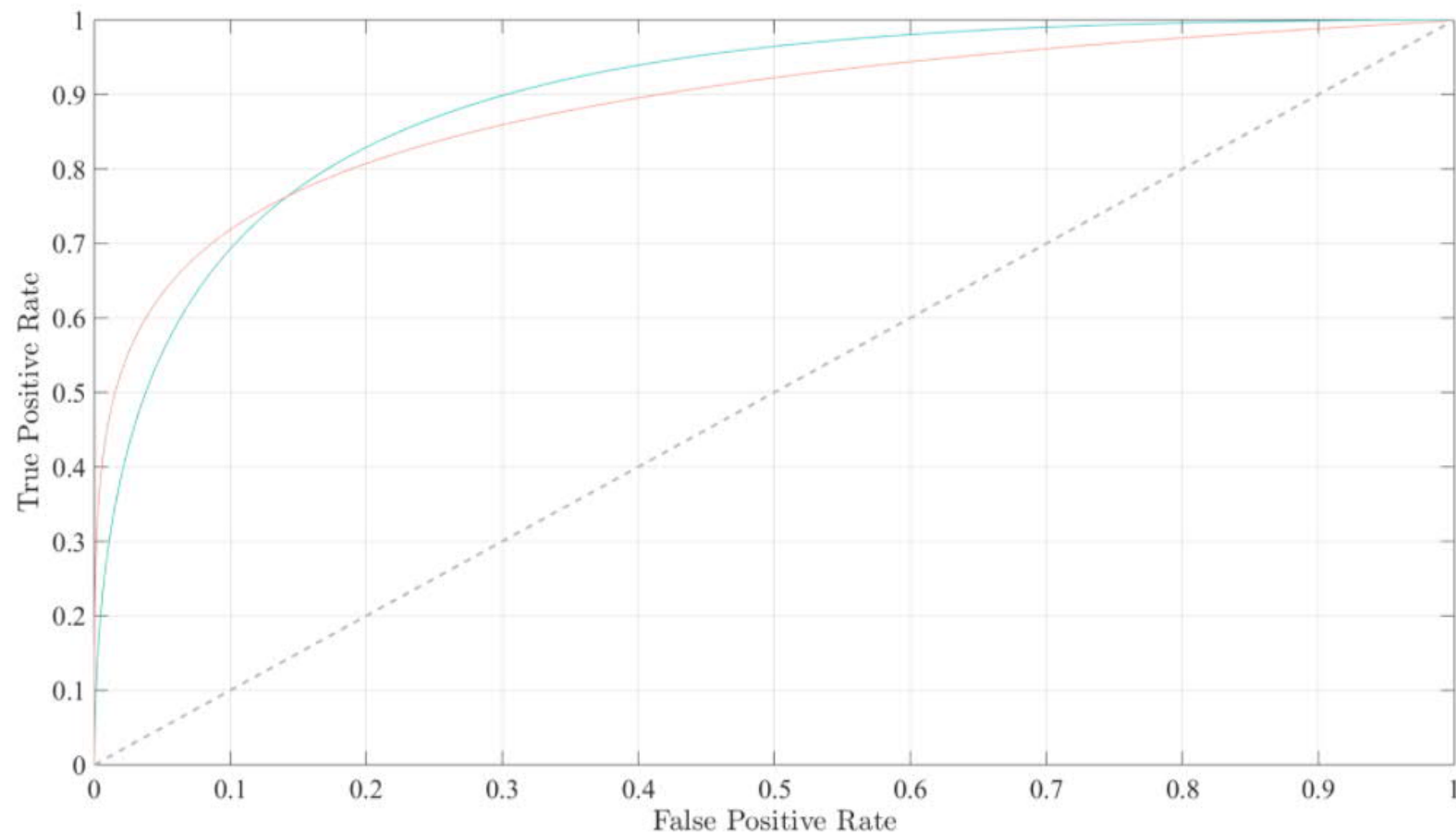
Test cases	Probability belong to positive class
e1	0.92
e2	0.90
e3	0.85
e4	0.85
e5	0.78
...	...



- ▶ Different probability thresholds lead to different classifiers
 - ▶ Each classifier has a point in the ROC space.

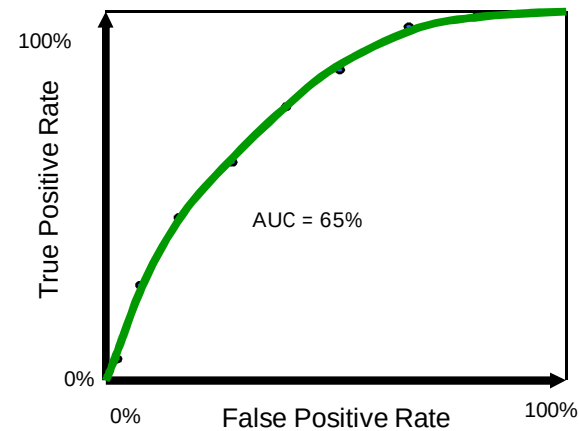
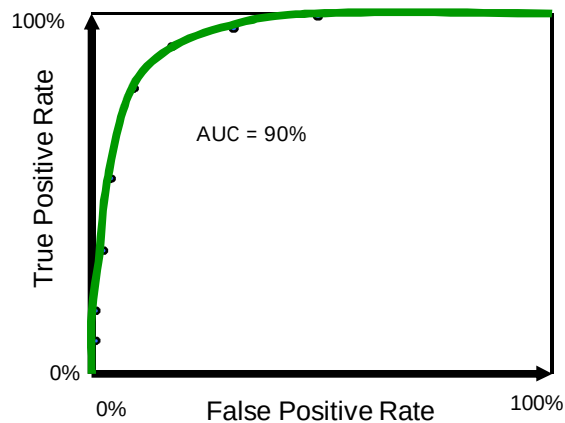
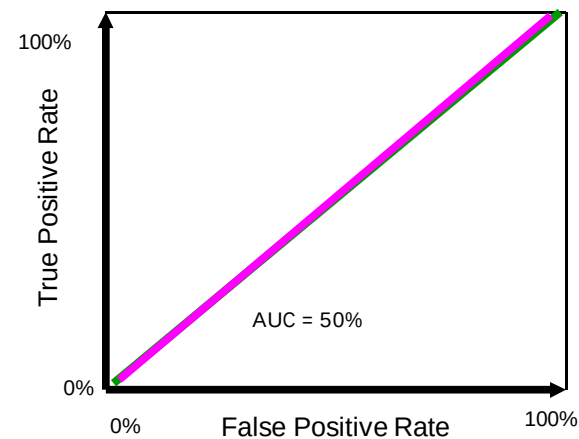
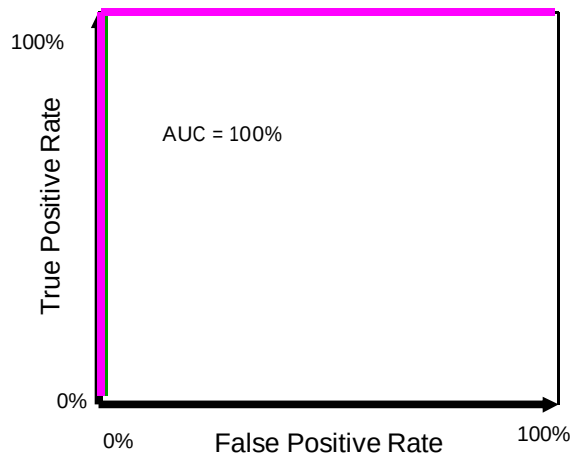
Comparing ROC Curves

- ▶ Which of the following two ROC curves is better?



AUC

- ▶ AUC = Area Under ROC Curve
- ▶ Overall measure of test performance



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 - ▶ Significance test

Performance Metrics for Regression

- ▶ **Mean Absolute Error (MAE)** computes the average deviation between predicted value y_i and the actual value $\hat{y}_i$ on a set of N examples:

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|$$

- ▶ **Mean Squared Error (MSE)** places more emphasis on larger deviations:

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

- ▶ **Root Mean Squared Error (RMSE)** has similar scale as MAE and places more emphasis on larger deviations:

$$RMSE = \frac{1}{N} \sqrt{\sum_{i=1}^N (y_i - \hat{y}_i)^2}$$

Performance Metrics for Regression (Cont'd)

- ▶ **Relative Absolute Error (RAE)** divides MAE by the MAE of a simple model that predicts the mean value:

$$RAE = \frac{\sum_{i=1}^N |y_i - \hat{y}_i|}{\sum_{i=1}^N |y_i - \bar{y}|}$$

- ▶ **Relative Squared Error (RSE)** divides MSE by MSE of a simple model that predicts the mean value:

$$RSE = \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}$$

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Performance Evaluation Methods

▶ Problem:

- ▶ Given a set S of data and a machine learning algorithm A , how do we evaluate the predictive performance of A on S ?
- ▶ We cannot learn a model from S and evaluate the model on S again because
 - ▶ Error on the training data is not a good indicator of a model's performance on future data.

▶ Evaluation methods

- ▶ Hold-out estimation
- ▶ Repeated hold-out estimation
- ▶ Cross validation
- ▶ Bootstrap

Performance Evaluation Methods

- ▶ Hold-out estimation
 - ▶ Randomly split the data set into a training set and a test set, e.g., training set (2/3), test set (1/3)
 - ▶ Build a model from the training set and estimate the error (or another measure, e.g., cost) on the test set
- ▶ Repeated hold-out estimation
 - ▶ Holdout estimate can be made more reliable by repeating the process with different random splits
 - ▶ For each split, an error rate is collected.
 - ▶ An overall error rate is obtained by averaging the error rates on the different splits

Performance Evaluation Methods (*Cont'd*)

- ▶ Cross-validation
 - ▶ Randomly divide the data set into k subsets of equal size
 - ▶ use $k-1$ subsets as training data and one subset as test data – do this k times using each subset in turn for testing
 - ▶ The error rates are averaged to yield an overall error estimate
 - ▶ This is called k -fold cross-validation

Performance Evaluation Methods (*Cont'd*)

- ▶ Stratified cross-validation
 - ▶ Ensures that classes in a subset have approximately the same distribution as in the original data set.
 - ▶ Stratification reduces the estimate's variance.
- ▶ Standard method for evaluation: stratified 10-fold cross-validation.
 - ▶ Why 10? Extensive experiments have shown that this is the best choice to get an accurate estimate.
 - ▶ There is also some theoretical evidence for this.

Performance Evaluation Methods (*Cont'd*)

▶ Leave-one-out

- ▶ A special case of k -fold cross-validation method when k equals to the number N of examples in the data set
- ▶ In each iteration, the test data set contains only one example and the training data set contains the rest $N-1$ examples.
- ▶ Advantages:
 - ▶ The greatest possible amount of data is used for training in each iteration.
 - ▶ Deterministic: no random sampling is involved
- ▶ Disadvantages:
 - ▶ Long running time
 - ▶ Non-stratified partitions of the data set

Evaluating Classifier Accuracy: Bootstrap

▶ Bootstrap

- ▶ Works well with small data sets
- ▶ Samples the given training examples randomly *with replacement*
 - ▶ i.e., each time an example is selected, it is equally likely to be selected again and re-added to the training set – resulting in a *bootstrap* sample

▶ Several bootstrap methods, and a common one is **.632 bootstrap**

- ▶ A data set with N examples is sampled N times, with replacement, resulting in a training set of N examples. The examples that did not make it into the training set end up forming the test set.
 - ▶ About 63.2% of the original data end up in the bootstrap, and the remaining 36.8% form the test set (since $(1 - 1/N)^N \approx e^{-1} = 0.368$)
- ▶ Repeat the sampling procedure k times, each leading to a model M_i . The overall accuracy of the model M :

$$Acc(M) = \frac{1}{k} \sum_{i=1}^k (0.632 \times Acc(M_i)_{test_set} + 0.368 \times Acc(M_i)_{train_set})$$

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 - ▶ *Significance Test*

Is the Difference between Two Models Significant?

- ▶ Are the cross-validation results of $Model_1$ and $Model_2$ on a data set S significantly different?
- ▶ Are the performances of $Model_1$ and $Model_2$ on a number of data sets significantly different?
- ▶ Paired t-test or other statistical tests can be used.

Paired t-test

- ▶ The *paired t-test* is a statistical hypothesis test that
 - ▶ tests the difference between the means for a pair of random samples. The observations are paired or matched in some way.
 - ▶ **Null hypothesis:** the two means are not significantly different

▶ Example:

Tree	Number of rusted leaves: year 1	Number of rusted leaves: year 2 (after treatment)
1	38	32
2	10	16
3	84	57
4	36	28
5	50	55
6	35	12
7	73	61
8	48	29
Average	46.8	36.2

Paired t-test (*Cont'd*)

- ▶ If you run a paired t-test on the example in the last slide, using a t-test program, say, at
 - ▶ <https://www.graphpad.com/quickcalcs/ttest2/>
- ▶ Result:
 - ▶ $t = 2.43$, degrees of freedom = 7
 - ▶ **p-value: 0.045**
 - ▶ The probability that the null hypothesis is true is 0.045
- ▶ If the p-value ≤ 0.05 , the null hypothesis can be rejected
 - ▶ Meaning the means of two samples are significantly different

How Paired t-test is Done Internally

► Calculate t-Statistic: $t = \frac{\bar{d}}{\frac{\sigma}{\sqrt{n}}}$

- $\bar{d}$ is the mean of the differences between the paired observations
- σ is the standard deviation of the differences
- n is the number of pairs

► Calculate the degree of freedom:

$$df = n-1$$

► Find the p-value:

- Find the critical t-value from the t-distribution table:
 - The value on the df row that is largest but smaller than the t-Statistic
- Get the corresponding p-value

► If the p-value ≤ 0.05 , the null hypothesis can be rejected

- Meaning the means of two samples are significantly different

	P						
one-tail	0.1	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	0.2	0.1	0.05	0.02	0.01	0.002	0.001
DF							
1	3.078	6.314	12.706	31.821	63.656	318.289	636.578
2	1.886	2.92	4.303	6.965	9.925	22.328	31.6
3	1.638	2.353	3.182	4.541	5.841	10.214	12.924
4	1.533	2.132	2.776	3.747	4.604	7.173	8.61
5	1.476	2.015	2.571	3.365	4.032	5.894	6.869
6	1.44	1.943	2.447	3.143	3.707	5.208	5.959
7	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	1.397	1.86	2.306	2.896	3.355	4.501	5.041
9	1.383	1.833	2.262	2.821	3.25	4.297	4.781
10	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	1.356	1.782	2.179	2.681	3.055	3.93	4.318
13	1.35	1.771	2.16	2.65	3.012	3.852	4.221
14	1.345	1.761	2.145	2.624	2.977	3.787	4.14
15	1.341	1.753	2.131	2.602	2.947	3.733	4.073

t-distribution table

Comparing Two Learning Algorithms on One Data Set

- ▶ Given two learning algorithms (L_1 and L_2) and a data set S , run k -fold cross-validation.

- ▶ Result of 10-fold:

Fold	Error rate (%) of L_1	Error rate (%) of L_2
1	5	7
2	2	1.9
3	7.8	5.7
4	4.9	5.1
5	12	15
6	8	9
7	7.6	9.8
8	10	11
9	6	5.9
10	8	9.8
Average	7.13	8.02

- ▶ Paired t-test result:

- ▶ p-value = 0.089

Comparing Two Learning Algorithms on a Number of Data Sets

- ▶ Given two learning algorithms (L_1 and L_2) and a few data sets $S_1, S_2, \dots, S_m$, run k -fold cross-validation on each algorithm and each data set.
- ▶ Results of 10-fold cross-validation of L_1 and L_2 on each data set:

Dataset	Average error rate (%) of 10-fold CV of L_1	Average error rate (%) of 10-fold CV of L_2
S_1	6.7	5.0
S_2	2	0.5
S_3	20.6	15.7
S_4	10.2	6.8
S_5	1.8	1.8
S_6	9	6.5
S_7	7.6	9.8
S_8	17	11
Average	9.36	7.14

- ▶ Paired t-test result:
 - ▶ p-value = 0.048

Note on Paired t-test

- ▶ When the sample size is less than 30, it is required that
 - ▶ Differences between a pair of samples are approximately normally distributed.
- ▶ If not normally distributed,
 - ▶ Wilcoxon signed-rank test can be used

Summary

- ▶ Predictive performance Measures
 - ▶ Classification
 - ▶ Accuracy, error, misclassification cost, precision, recall, F-score, AUC
 - ▶ Regression
 - ▶ MAE, MSE, RMSE, RAE, RSE
- ▶ Performance Evaluation Methods
 - ▶ Holdout
 - ▶ Repeated holdout
 - ▶ Cross-validation
 - ▶ Bootstrap
- ▶ Methods for model comparisons
 - ▶ Significance Test